

Probabilistic Classification

CSC 487: Deep Learning
Instructor: Jonathan Ventura

Probabilistic classifier

- Define z to be the **log-odds** of the positive class

$$z = \log \frac{P(y = 1 \mid \mathbf{x})}{1 - P(y = 1 \mid \mathbf{x})}$$

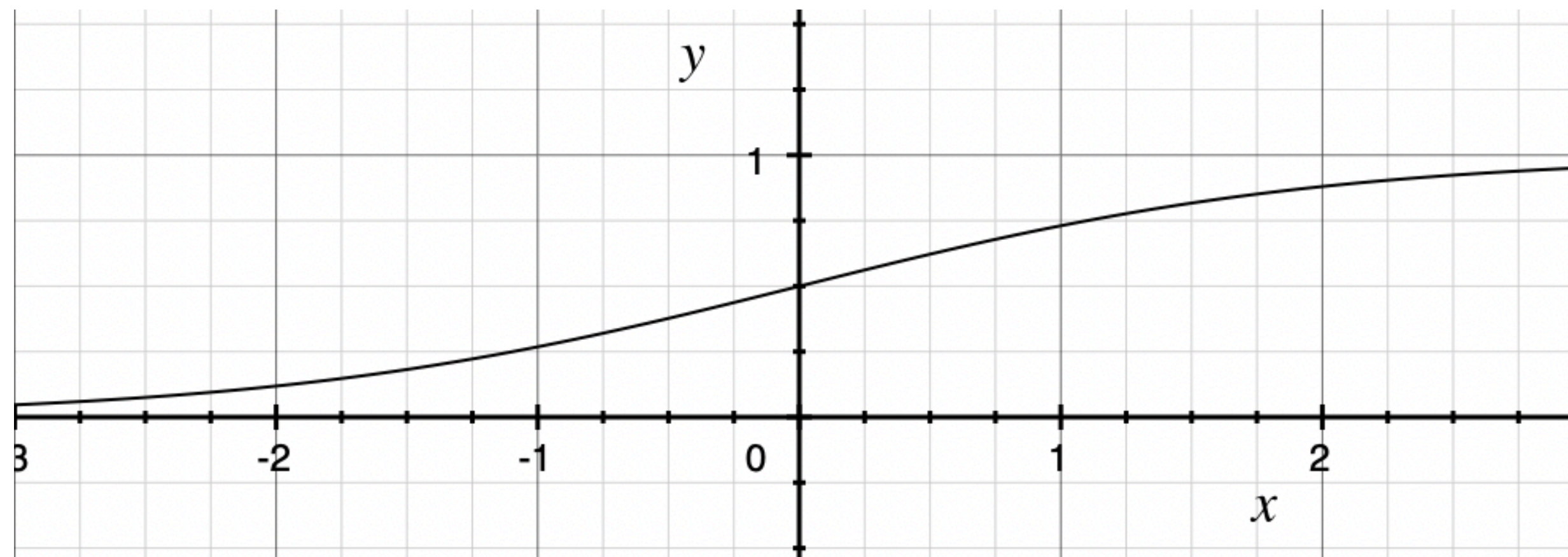
- Conditional probability of positive class is a **logistic function** of the linear predictor z :

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-z}}$$

What is the z value?

- Conditional probability of positive class is a **logistic function** of the linear predictor z :

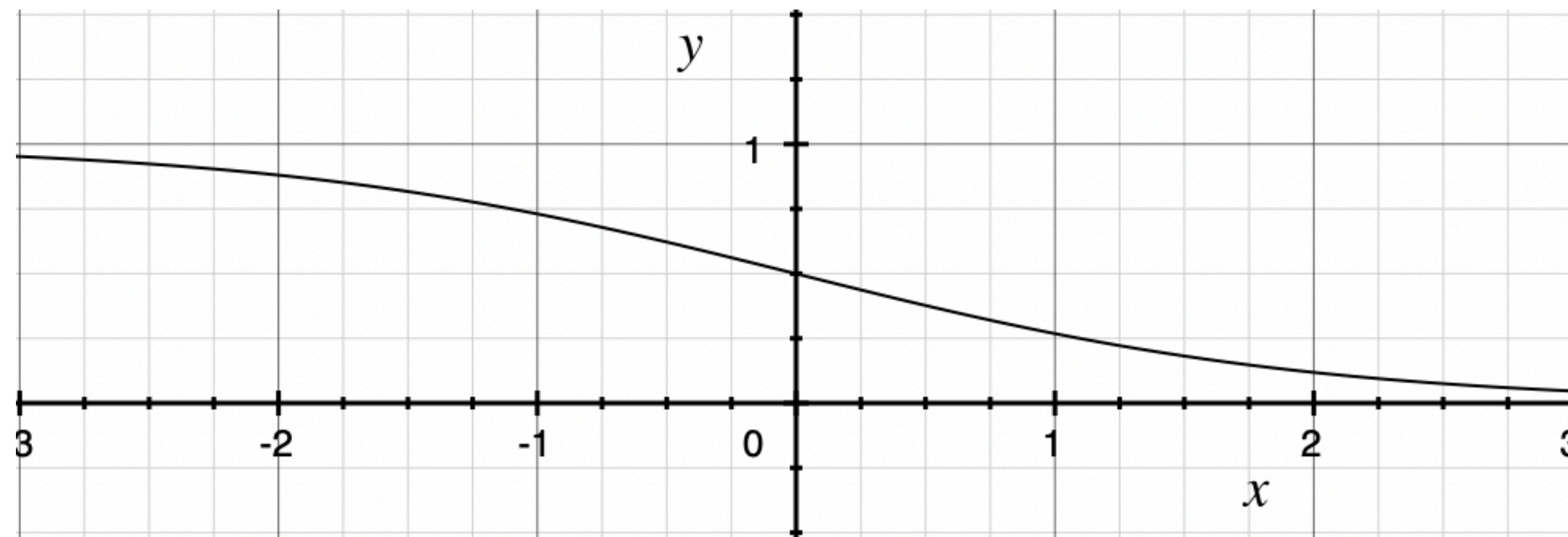
$$P(y = 1 | \mathbf{x}) = \frac{1}{1 + e^{-z}}$$



What is the z value?

- Conditional probability of positive class is a **logistic function** of the linear predictor z :

$$P(y = 1 | \mathbf{x}) = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$



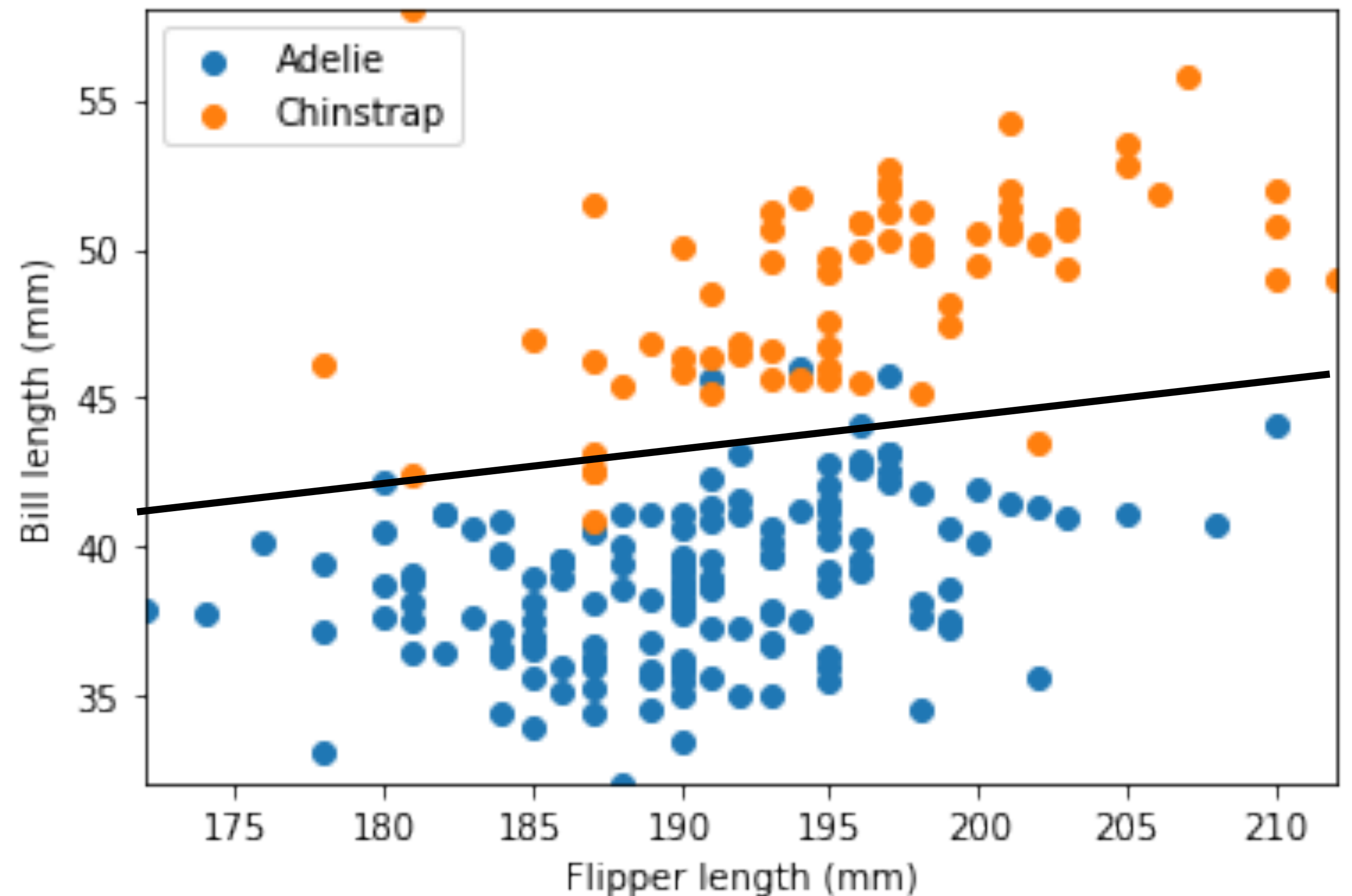
Learning the optimal hyperplane

- Given a set of labeled points, how do we find the optimal hyperplane to separate the two classes?

- Idea:

- Maximize** the probability of the correct class for each point

$$\arg \max \prod_i P(y^i | \mathbf{x}^i)$$



Learning the optimal hyperplane

- **Maximize** the probability of the correct class for each point

$$\arg \max \prod_i P(y^i | \mathbf{x}^i)$$

- Easier to maximize the **log-likelihood**:

$$\arg \max \sum_i \log P(y^i | \mathbf{x}^i)$$

- Equivalent: minimize the **negative log-likelihood**:

$$\arg \min \sum_i -\log P(y^i | \mathbf{x}^i)$$

Learning the optimal hyperplane

- What is the negative log likelihood?

$$-\log P(y^i | \mathbf{x}^i) = \begin{cases} -\log \frac{1}{1 + e^{-z^i}} & \text{if } y^i = 1 \\ -\log \frac{1}{1 + e^{z^i}} & \text{if } y^i = 0 \end{cases}$$

Learning the optimal hyperplane

- What is the negative log likelihood?

$$-\log P(y^i | \mathbf{x}^i) = \begin{cases} \log(1 + e^{-z^i}) & \text{if } y^i = 1 \\ \log(1 + e^{z^i}) & \text{if } y^i = 0 \end{cases}$$

Learning the optimal hyperplane

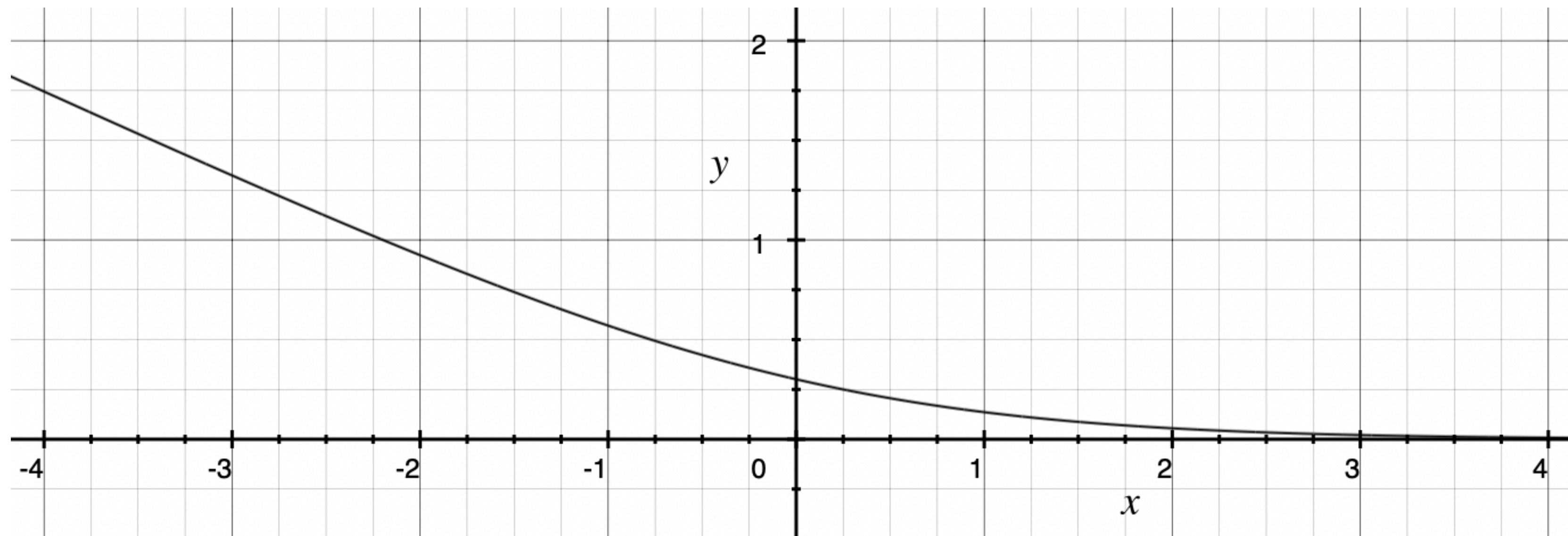
- What is the negative log likelihood?

$$-\log P(y^i | \mathbf{x}^i) = y^i \log(1 + e^{-z^i}) + (1 - y^i) \log(1 + e^{z^i})$$

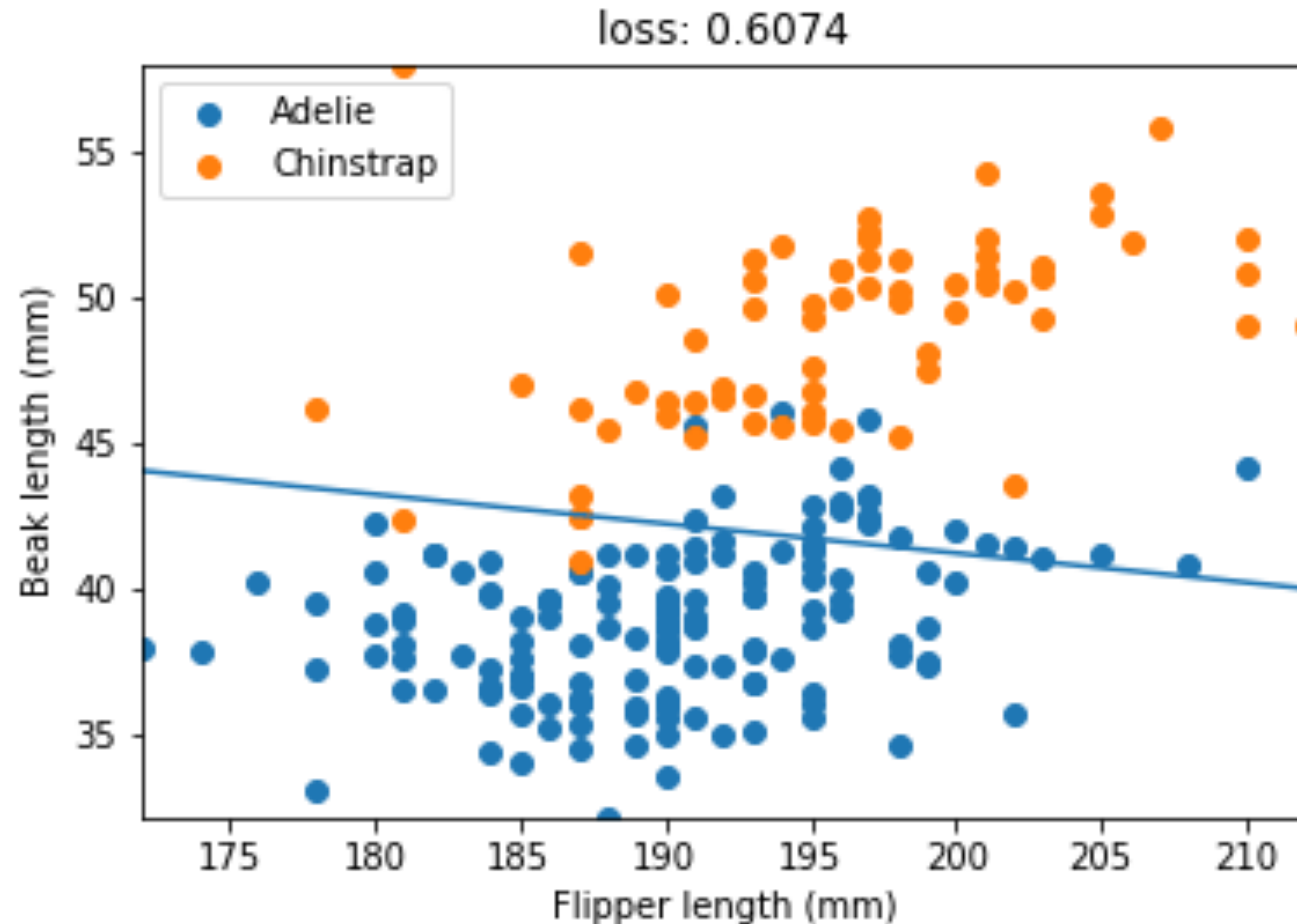
Learning the optimal hyperplane

- The negative log likelihood is our “loss” function:

$$L^i = y^i \log(1 + e^{-z^i}) + (1 - y^i) \log(1 + e^{z^i})$$



Penguins Dataset Example



Each point is assigned to the classifier with greatest z value

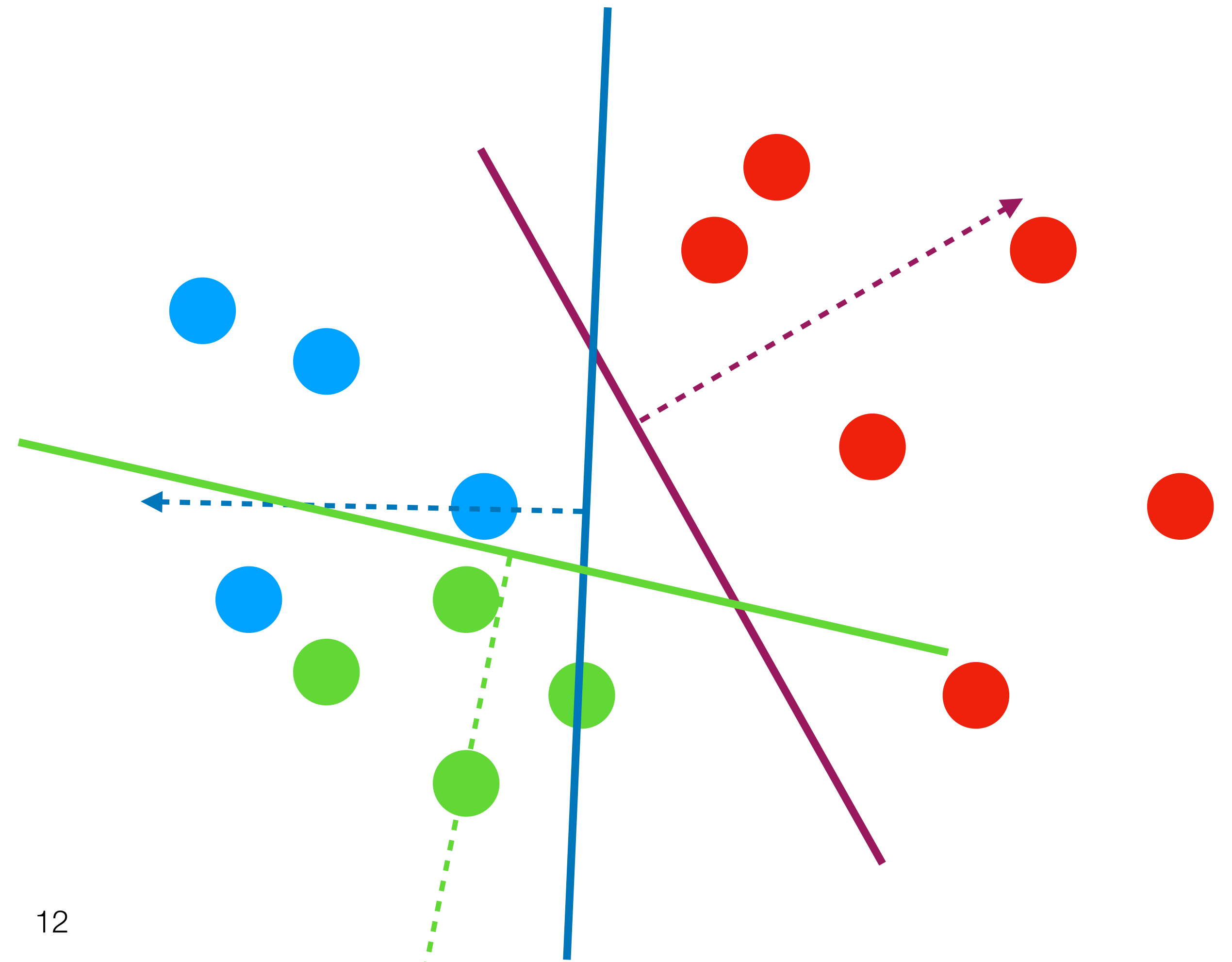
- The z values are called “logits”

$$z_0 = \mathbf{w}_0^T \mathbf{x} + b_0$$

$$z_1 = \mathbf{w}_1^T \mathbf{x} + b_1$$

$$z_2 = \mathbf{w}_2^T \mathbf{x} + b_2$$

$$y = \operatorname{argmax}_j z_j$$



Probability model for multi-class classification

- Conditional probability of class y_i :

$$P(y^i | \mathbf{x}^i) = \frac{e^{z_{y_i}^i}}{\sum_j e^{z_j^i}}$$

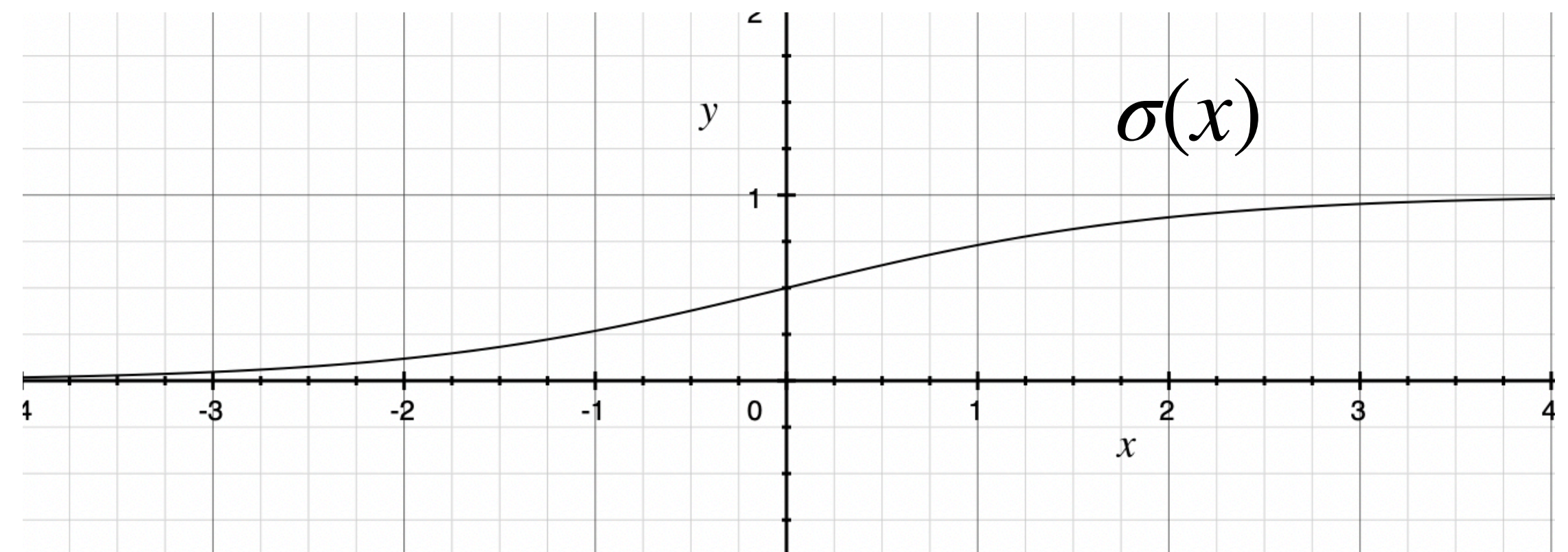
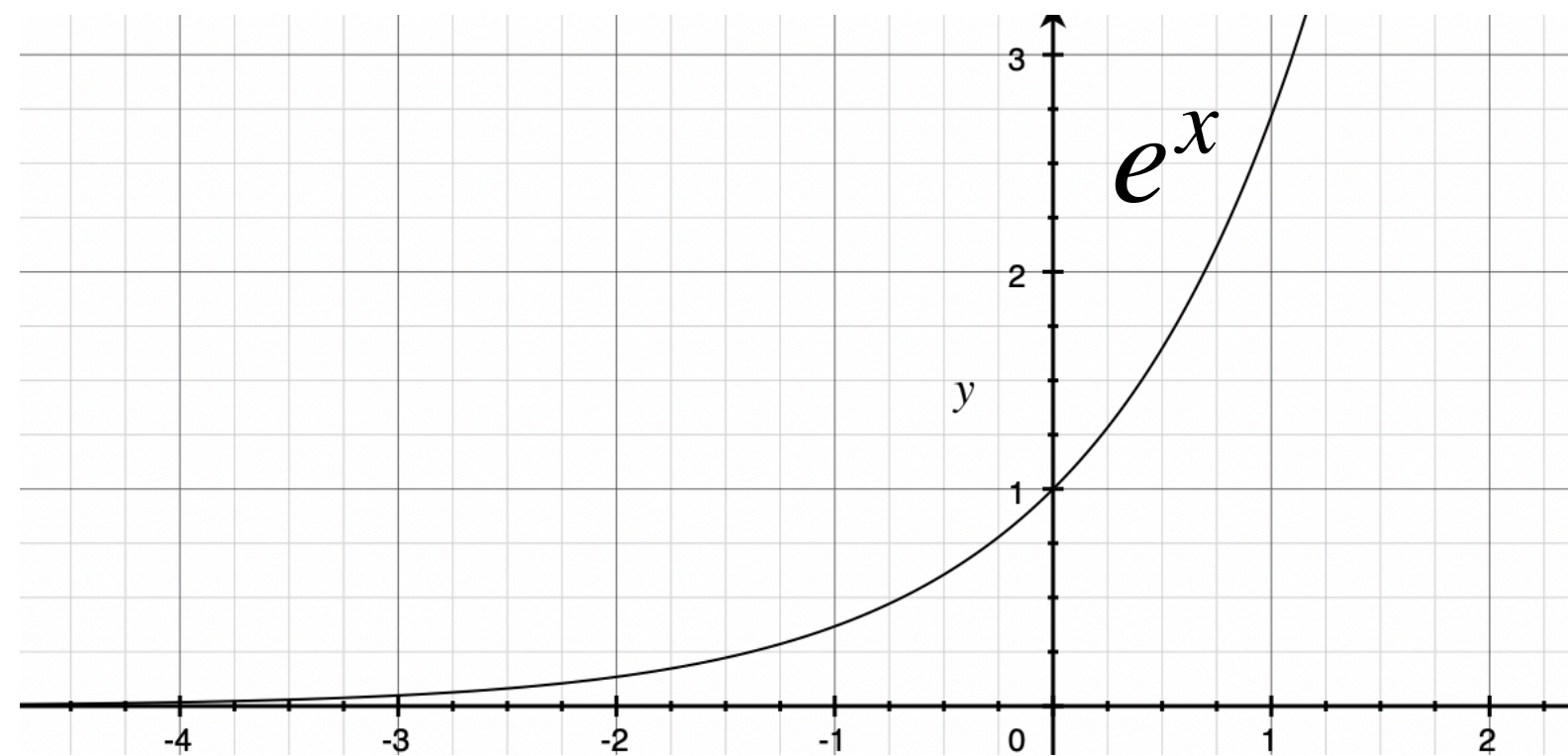
- This is called the “softmax” function.

Simple binary classifier

- Softmax ensures that output values are positive and sum to one:

$$\sigma(z_0) = \frac{e^{z_0}}{e^{z_0} + e^{z_1}}$$

$$\sigma(z_1) = \frac{e^{z_1}}{e^{z_0} + e^{z_1}}$$



Loss function for multi-class classification

- Negative log-likelihood:

$$-\log P(y^i | \mathbf{x}^i) = -\log e^{z_{y_i}^i} + \log \sum_j e^{z_j^i}$$

$$= \underbrace{-z_{y_i}^i}_{\text{Increase likelihood of correct class}} + \log \underbrace{\sum_j e^{z_j^i}}_{\text{Decrease likelihood of all classes}}$$

Increase likelihood of correct class

Decrease likelihood of all classes

Penguins Dataset

